Integer Natural Evolution Strategies

A lattice-native natural-gradient evolution strategy for integer optimization

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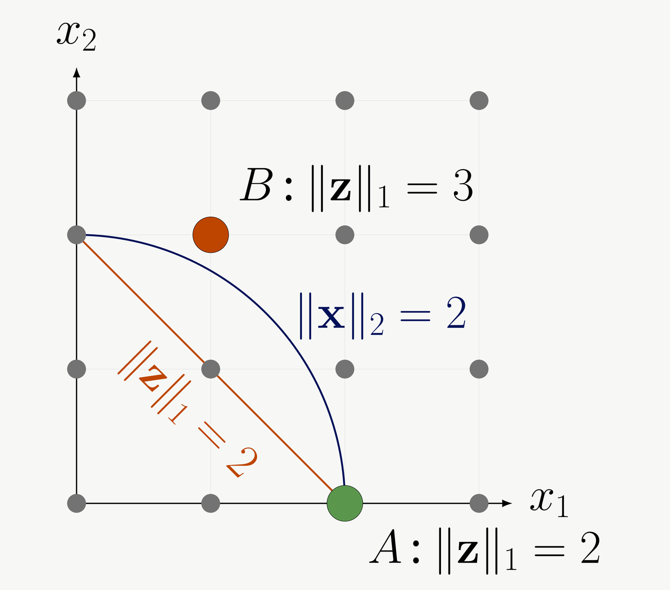
Summary

- Gaussian mutations are natural for continuous ESs; when turning to unbounded integer spaces, **Double-Geometric (DG) mutations obtain maximal entropy** [1].
- **Natural Evolution Strategies** provide a principled way to adapt parameters.
- The ℓ_1-norm is the natural measure of displacement on $\mathbb{Z}^n$, and the DG distribution is its canonical mutation operator [2]. The sufficient statistic for this family gives a natural-gradient signal for **step-size adaptation**.
- Cumulating this signal through an evolution path yields the $(1,\lambda)$-INES.
- INES learns meaningful coordinate-wise search scales and performs particularly well in **higher dimensions**, especially on Ellipse-like landscapes.

Research question. *Can integer step-size adaptation be derived from first principles using a native ℓ_1-norm mutation model and information geometry?*

Why ℓ_1 on the integer lattice?



Rounding points from the **same Euclidean shell** can produce lattice steps with different ℓ_1 lengths.

$$\|\mathbf{x}\|_2 = d \quad \not\Rightarrow \quad \|\mathbf{z}\|_1 = d$$

with

$$\mathbf{z} = \mathrm{rnd}(\mathbf{x}).$$

For $d = 2$; A has ℓ_1 length 2, while B has ℓ_1 length 3.

Key point: Euclidean radius does not uniquely determine integer step length.

A lattice-native mutation model

The **Double-Geometric (DG)** distribution is the maximum-entropy mutation model on $\mathbb{Z}$ for a fixed expected absolute displacement [1]:

$$\Pr(Z = z) = \frac{1-q}{1+q}\, q^{|z|}, \qquad z \in \mathbb{Z}.$$

We therefore parameterize mutation strength directly by

$$\boxed{\delta_i := \mathbb{E}[|Z_i|]}, \qquad \delta_i = \frac{2q_i}{1-q_i^2}.$$

The DG is an exponential family with

$$\theta_i = \log q_i, \qquad \boxed{\phi(z_i) = |z_i|}.$$

Consequently,

$$\mathbb{E}[\phi(Z_i)] = \mathbb{E}[|Z_i|] = \delta_i, \quad \text{and,} \quad \mathrm{Var}[\phi(Z_i)] = \mathrm{Var}[|Z_i|] = \delta_i\sqrt{1+\delta_i^2}.$$

Key observation: $|Z_i|$ plays two roles at once: it measures lattice displacement and is the **sufficient statistic** for the DG dispersion.

The sufficient statistic gives the adaptation signal

For an exponential family, the likelihood score is the **observed sufficient statistic minus its expectation**. For the DG family, the Fisher information is $\mathrm{Var}[|Z_i|]$.

Selected step $|z_{s,i}|$ → **Center** $|z_{s,i}| - \mathbb{E}[|Z_i|]$ likelihood score → **Normalize** $\dfrac{|z_{s,i}| - \mathbb{E}[|Z_i|]}{\mathrm{Var}[|Z_i|]}$ natural gradient

Using the DG moments,

$$\boxed{\widetilde{\nabla}_{\theta_i} = \frac{|z_{s,i}| - \delta_i}{\delta_i\sqrt{1+\delta_i^2}}}$$

Interpretation: compare the selected step with its expectation, then **normalize by the Fisher information**.

Can we accumulate the signal?

Yes. Fading memory is well motivated for online natural-gradient learning [3]; here, it provides a momentum-like adaptation of the natural-gradient signal.

$$\pi_i \leftarrow (1-c)\pi_i + \sqrt{c(2-c)}\, g_i, \qquad g_i = \frac{|z_{s,i}| - \delta_i}{\max\{\delta_i\sqrt{1+\delta_i^2}, 1\}}.$$

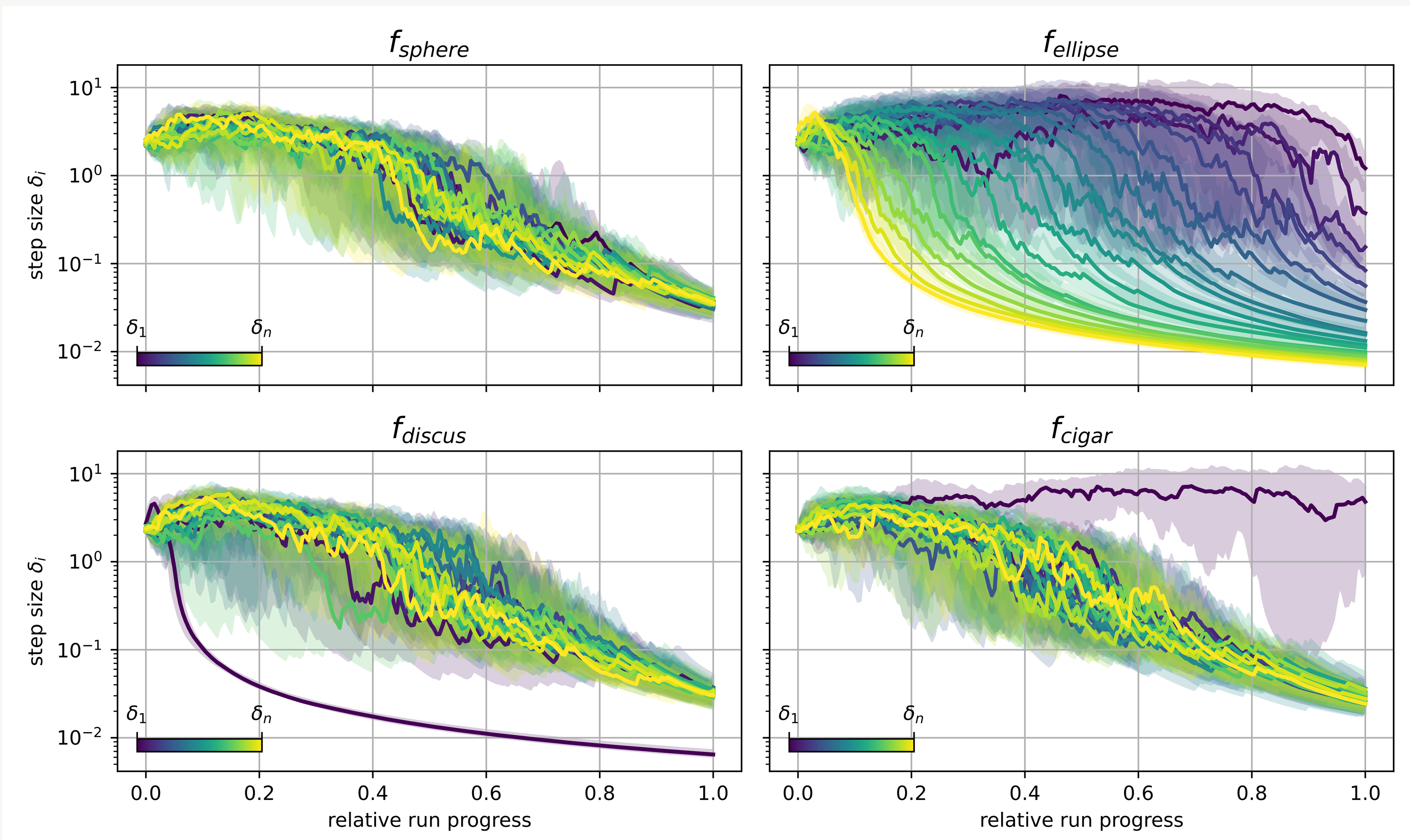
We **prove** that this update is a fading-memory online natural gradient estimator.

A concrete $(1,\lambda)$-INES

1. For $k = 1, \ldots, \lambda$: sample $\mathbf{z}_k \sim \bigotimes_i \mathrm{DG}(q_i(\delta_i))$ and evaluate $f(\mathbf{m} + \mathbf{z}_k)$.
2. Select $s = \arg\min_k f(\mathbf{m} + \mathbf{z}_k)$ and set $\mathbf{m} \leftarrow \mathbf{m} + \mathbf{z}_s$.
3. Compute the Fisher-normalized signal g_i and update the cumulative path π_i.
4. Adapt the expected absolute step size: $\boxed{\delta_i \leftarrow \delta_i \cdot \exp(\eta\pi_i)}$.

Calibration in the paper: $\lambda = 10$, $c(n) = 1 - 1.5/n$, and $\eta(n) = (2/n)^{1/3}$.

INES learns the coordinate structure

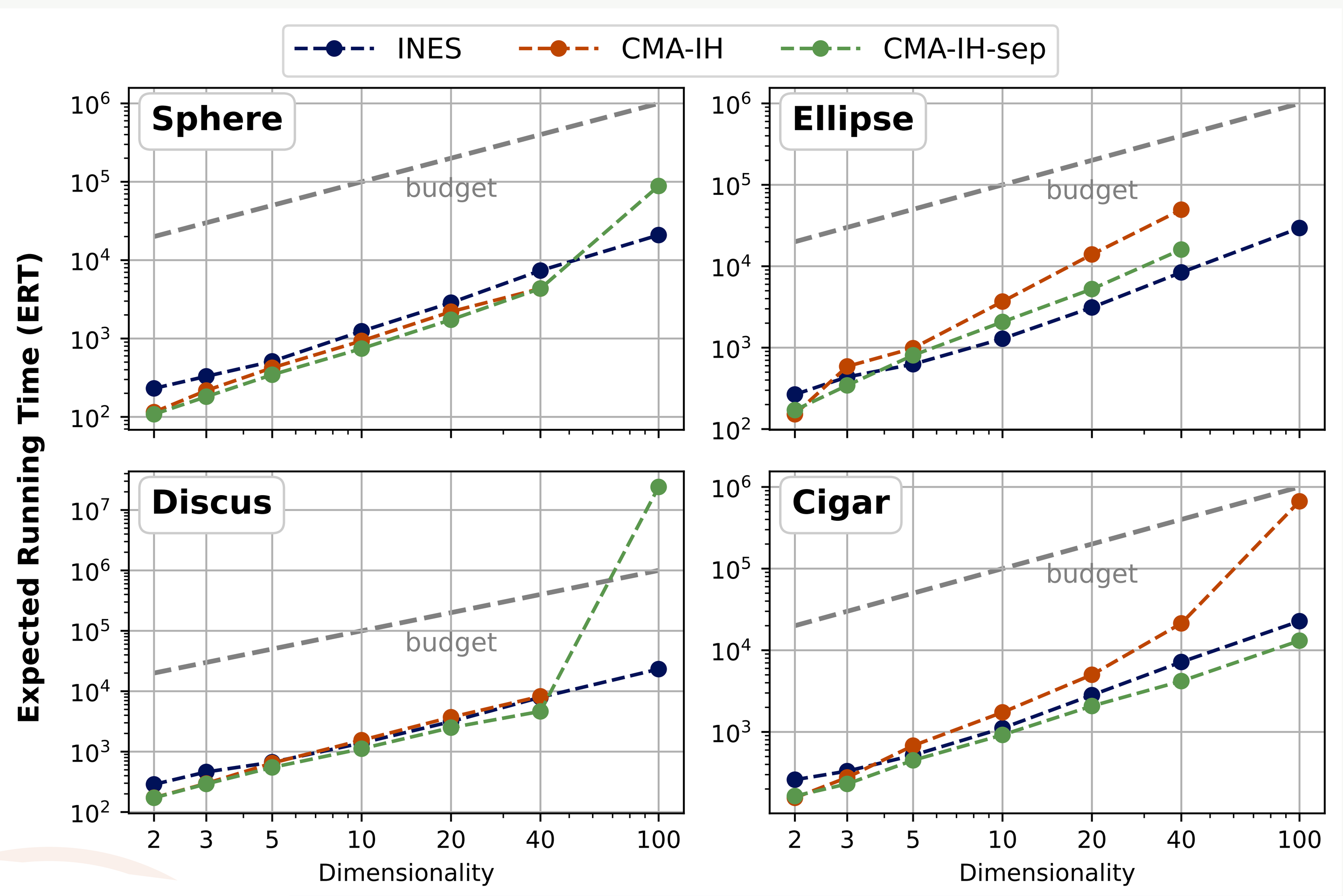


Sphere: coordinates evolve similarly.
Discus: the distinguished coordinate is suppressed.
Ellipse: step sizes separate according to conditioning.
Cigar: the distinguished coordinate is amplified.

The learned step sizes behave as expected from the benchmark geometry, supporting the interpretation of δ_i as meaningful coordinate-wise scales.

Benchmarking on Unimodal Quadratic functions

25 runs per function/dimension, budget $10^4 \cdot n$; lower ERT is better.



Conclusions & future work

- The DG distribution gives a **lattice-native** mutation model for integer ES.
- A natural-gradient signal for step-size adaptation is provided by the sufficient statistic.
- Cumulation turns this signal into a stable $(1,\lambda)$-INES update.
- INES is most compelling on **Ellipse** and remains robust in high-dimensional settings.

Current limitations / next steps

No explicit global step-size component; coordinate-wise adaptation only. Natural extensions are multi-parent recombination, a global scale, and modeling dependencies between integer variables.

References

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Paper
Code

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